MARS BIOSPHERE COLLAPSE VS. EMERGENCE

A Pre-Registered Geometric Prediction of the Complexity Gradient in the Mars Sedimentary Record Prior to Sample Return, the Water Attractor State Cascade Model, and the Two-Axis LECA Stabilisation Framework

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Repository: github.com/Eric-Robert-Lawson/attractor-oncology

The Cheyava Falls fossil record (Hurowitz et al., *Nature* 645, 2025) is consistent with either the beginning or the end of a Martian biosphere. These two interpretations make opposite predictions about the complexity gradient in the Mars subsurface sedimentary record: if Cheyava Falls represents biosphere *emergence*, complexity increases with decreasing depth; if it represents biosphere *collapse*, complexity increases with increasing depth and age. This prediction is derived from geometric first principles, is locked here before Mars sample return, and is testable by the Mars sample return mission (2030s target). Pre-registered: 2026-03-13.

Abstract

We present a pre-registered geometric prediction concerning the directional ambiguity in the Cheyava Falls biosignature record on Mars. The magnetic field loss of Mars (∼4.0–4.1 Ga) and the Cheyava Falls deposition age (∼3.8–4.1 Ga) are coincident within measurement error, creating a fundamental directional ambiguity: the fossil may represent life establishing itself (emergence) or life in terminal decline (collapse). We derive two mutually exclusive predictions from attractor geometry that are testable by the complexity gradient in the Mars subsurface sedimentary record, to be resolved by Mars sample return. We additionally present the water attractor state cascade model, which identifies three distinct chemical attractor states of planetary water chemistry and the biological grades each supports. We present the two-axis LECA stabilisation model, which identifies independent nitrogen and oxygen destabilisation axes that together define the conditions under which the LECA attractor is the stable state of carbon chemistry. All derivations are from geometric first principles. This document is the pre-registration of the complexity gradient prediction before the Mars sample return mission retrieves material from the Jezero Crater sedimentary record.

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1. The Timestamp Collision

Two facts are established independently of this framework:

(1) The Mars magnetosphere was lost approximately 4.0–4.1 Ga, as determined by MAVEN mission crustal magnetisation measurements.

(2) The Cheyava Falls biosignature features in the Bright Angel formation of Jezero Crater are

dated approximately 3.8–4.1 Ga, as reported in Hurowitz et al. [5].

These ranges overlap within measurement error. The magnetic field loss and the depositional age of the fossil features occurred within the same geological event window.

This creates a directional ambiguity that cannot be resolved from surface data alone.

The fossil record has no direction arrow. It records what was present. It does not record whether what was present was growing or dying.

2. The Two Interpretations

2.1. Option A — Emergence

Under this interpretation, the Cheyava Falls lake represents a newly habitable environment. Life was forming or recently established in State 1 water conditions. The LUCA attractor was establishing in a new depositional environment. The fossil is the beginning of a cascade that was then interrupted by atmospheric collapse.

Complexity gradient prediction (Option A): Biosignature complexity *decreases* with increasing depth and age. Simpler organisms deeper and earlier. More complex features shallower and later. Cheyava Falls is the most developed record in the sequence.

2.2. Option B — Collapse

Under this interpretation, Mars harboured a more complex biosphere that predated Cheyava Falls. The magnetic field loss initiated atmospheric stripping. Conditions deteriorated progressively. Complex organisms, requiring the most stable conditions, died first. Simpler chemotrophic organisms survived longest. LUCA-grade iron-sulfur chemotrophs were the last surviving life form. The Cheyava Falls leopard spots are the final record of a dying world.

Complexity gradient prediction (Option B): Biosignature complexity *increases* with increasing depth and age. More complex features deeper and earlier. Simpler features shallower and later. Cheyava Falls is the least developed record in the sequence — the last survivor.

2.3. Why Current Data Cannot Decide

The surface record is the survival record, not the complete record.

Three billion years of unshielded solar and cosmic ray bombardment (post-magnetosphere loss at 4.0–4.1 Ga) preferentially destroys complex organic biosignatures. Simple iron-sulfur minerals — vivianite, greigite — survive. Complex organic molecules, proteins, and nucleic acids do not.

Whatever complexity existed at Cheyava Falls at time of deposition has been partially or entirely degraded by radiation.

The surface record is the floor of complexity, not the ceiling.

Additionally, Mars never underwent a Great Oxidation Event (GOE) equivalent. The entire water column remained anoxic throughout the habitable period. This produces colony-scale mineral reaction fronts as the dominant fossilisation mechanism rather than individually resolved cellular fossils (see Section 5).

Both Option A and Option B are consistent with the currently available surface data.

3. The Pre-Registered Prediction

Pre-Registered Prediction (locked 2026-03-13, prior to Mars sample return):

The complexity gradient in the Jezero Crater sedimentary record with depth and increasing age will run in one of two directions:

If collapse (Option B): Subsurface samples at greater stratigraphic depth than the Cheyava Falls surface sample will show *more* complex biosignatures, specifically:

- More diverse mineral assemblages.
- Richer and more varied organic carbon content.
- More complex spatial organisation of mineral–organic associations.
- Biosignature features at smaller scale (approaching individual cell scale of 10–30 μm rather than colony scale of 1–2 mm) if preservation conditions permit.

If emergence (Option A): Subsurface samples at greater stratigraphic depth will show *less* complex biosignatures, consistent with earlier, simpler biological communities.

The Mars sample return mission resolves which prediction is correct.

4. The Water Attractor State Cascade Model

4.1. Water Chemistry as Attractor State

Water is not a uniform medium. The chemical composition of a body of water constitutes an attractor state of the water chemistry itself. Different attractor states support different grades of biological complexity. The cascade from first life to a biologically maintained atmosphere is a cascade of coupled water attractor states.

4.2. The Three States

State	Chemistry	Life supported	Transition
State 1	pH 5–6; high Fe^{2+}; anoxic; low N; pre-GOE	LUCA-grade chemolithotrophs only. Cannot support stable LECA.	Biological photosynthesis drives GOE.
State 2	pH 6.5–7.5; Fe^{2+} declining; O_2 rising; N cycling increasing	LECA-grade eukaryotes. Mitochondria functional. LECA attractor stable.	LECA diversification drives State 3.
State 3	pH 8.1; trace Fe; high sulfate; O_2 abundant	Full triadic cycle. Biologically maintained atmosphere.	Terminal state on Earth.

Table 1: The three water attractor states on Earth. Each transition is driven by biological activity in the previous state.

4.3. Mars in This Framework

The Cheyava Falls mineralogy (vivianite = $Fe_3(PO_4)_2$; greigite = Fe_3S_4; anoxic conditions; organic carbon) confirms State 1 water chemistry in the Jezero Crater lake environment [5].

Mars had State 1 water. The magnetic field loss at 4.0–4.1 Ga prevented oxygen accumulation. No GOE equivalent occurred. The cascade was interrupted at State 1. Mars never reached State 2.

5. The Two-Axis LECA Stabilisation Model

5.1. Two Independent Destabilisation Axes

The LECA attractor is stabilised by the *simultaneous* absence of two independent destabilisation signals.

Axis 1 — Nitrogen: Bioavailable nitrogen (NH_4^+, NO_3^-, or amino acids above threshold) fires germination and multicellular organisation programs. In the LECA arrest medium: explicitly absent. Tested in ARM A by nitrogen release control at 72 hours (addition of $(NH_4)_2SO_4$ restores germination).

Axis 2 — Oxygen: Sufficient O_2 (above ∼3% for microaerophilic threshold) activates aerobic metabolic programs unavailable in pre-GOE conditions. Mitochondria at full aerobic capacity provide the energy density to advance developmental programs past D=0. In the LECA arrest medium: held at <3% O_2 throughout.

5.2. What the GOE Did

On Earth, the GOE (∼2.4 Ga) raised Axis 2 above the destabilisation threshold. Concurrent nitrogen cycling changes raised Axis 1. Both axes exceeded their thresholds together. The LECA attractor was destabilised at planetary scale. Complex eukaryotic life emerged.

On Mars, the magnetic field loss prevented oxygen accumulation. Axis 2 never rose above threshold. Nitrogen cycling could not change. Axis 1 remained below threshold. The LECA attractor remained stable in an increasingly hostile and ultimately uninhabitable planetary environment.

5.3. Impact on the Arrest Medium

This analysis confirms that the LECA Resurrection Protocol arrest medium (pre-registered DOI: 10.5281/zenodo.18986790) correctly addresses both axes. No change to the protocol is required. The two-axis understanding deepens the mechanistic explanation of why the conditions are specified as they are.

6. The Fossilisation Paradox and Its Resolution

On Earth, the best cellular fossils are preserved at redox boundaries between an oxidising surface layer and anoxic deep sediment. This produces sharp mineralisation of organic material at the boundary, sometimes preserving cellular detail.

Mars never had a GOE equivalent. The entire water column remained anoxic. No sharp oxidising–anoxic boundary existed.

The predicted fossilisation mechanism for State 1 water conditions is therefore:

- Not individual cellular-detail fossils.
- Colony-scale mineral reaction fronts.
- The bulk mineral product of microbial community metabolism at population scale.

- Vivianite and greigite formed by Fe-phosphate and Fe-sulfide reducing metabolism at 1–10 mm scale (colony scale) rather than 10–30 μm scale (individual cell scale).

This is not a failure of the biological interpretation of Cheyava Falls. It is the geometric prediction of what biosignatures look like on a world that never underwent a GOE.

The leopard spots (1–2 mm, vivianite–greigite) are consistent with this mechanism. The poppy seeds (<200 μm) approach colonial microstructure scale. Individual cellular-scale confirmation (10–30 μm) requires Mars sample return resolution.

7. The ARM A Reference Standard for Sample Return

The LECA Resurrection Protocol ARM A (pre-registered DOI: 10.5281/zenodo.18986790, timestamp: 2026-03-12) will produce a formally defined, pre-registered morphological profile of a living LECA-grade cell.

This profile constitutes the formal reference standard against which Mars sample return material should be compared.

The ARM A reference standard:

- Is pre-registered before the Mars samples arrive on Earth.
- Is derived from the same geometric framework that generated the Cheyava Falls prediction.
- Is physically anchored by laboratory confirmation in living biological material.
- Formally defines the expected cellular diameter (10–30 μm), nuclear morphology (DAPI-stained nucleus confirmed), and validation criteria for LECA-grade cell identification.

If the Cheyava Falls sample return material contains sub-features at 10–30 μm scale matching the ARM A morphological profile, that constitutes individual cell-scale confirmation of LECA-grade life on ancient Mars.

The relationship between ARM A (~$200, BSL-1, days to result) and Mars sample return (multi-billion dollar mission, 2030s) is explicit: the cheap experiment in Rogers Park, Chicago provides the calibration standard for the most consequential planetary science mission in human history.

8. Summary of Pre-Registered Predictions

#	Prediction	Test	Resolves
P1	Complexity gradient increases with depth if collapse (Option B).	Mars sample return. Jezero Crater subsurface.	Whether Cheyava Falls is a death record or a birth record.
P2	Complexity gradient decreases with depth if emergence (Option A).	Mars sample return.	Same.
P3	Sub-features at 10–30 μm in the poppy seeds matching the ARM A profile confirm individual LECA-grade cells.	Mars sample return nanometre-resolution imaging.	Individual cell-scale confirmation of LECA-grade life on ancient Mars.

Table 2: Pre-registered predictions locked 2026-03-13, prior to Mars sample return.

9. Honest Statement of Limitations

(1) **Surface data is insufficient.** The collapse vs. emergence distinction cannot be resolved from Perseverance rover surface data. It requires access to the stratigraphic depth record in the Cheyava Falls formation. Mars sample return provides this.

(2) **Radiation taphonomy is uncertain.** 3.8 billion years of unshielded radiation have preferentially degraded complex biosignatures. Even under Option B (collapse), the subsurface record may not preserve sufficient complexity contrast to distinguish the two interpretations. Radiation shielding at depth is the critical variable.

(3) **ARM A pending.** The morphological reference standard requires ARM A confirmation. Until ARM A runs, the reference standard is specified but not physically anchored.

(4) **Common origin vs. independent emergence unresolved.** This document does not address whether any Martian life shared common ancestry with Earth life. Both scenarios are consistent with the geometric prediction. The distinction requires genomic analysis of Mars sample return material.

10. Conclusion

The Cheyava Falls fossil record is ambiguous in direction. The ambiguity is geometric, not observational, and cannot be resolved from currently available data.

Two mutually exclusive predictions follow from attractor geometry:

- Under biosphere collapse (Option B): complexity increases with depth.
- Under biosphere emergence (Option A): complexity decreases with depth.

This prediction is locked here, on 2026-03-13, before the Mars sample return mission retrieves subsurface material from Jezero Crater.

The water attractor state cascade model and the two-axis LECA stabilisation model are presented as the geometric basis for understanding why State 1 water conditions on ancient Mars were the conditions in which the LUCA attractor was the stable biological state.

ARM A provides the morphological reference standard that makes the Mars sample return comparison formally defined rather than interpretively open.

The geometry was derived before the data.

The data will confirm which geometric prediction is correct.

Both outcomes are informative.

Neither can be adjusted after the fact.

This document is the lock.

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This work was conducted independently without institutional funding or affiliation. The geometric derivation sequence is documented in the pre-registration record (DOI: 10.5281/zenodo.18986790) and the attractor-oncology repository (github.com/Eric-Robert-Lawson/attractor-oncology). The Cheyava Falls data is the work of the NASA Perseverance mission science team (Hurowitz et al., *Nature*, 2025) and is referenced here as independent contextual data, not as a derivation source.

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The geometry came first. The data will say which prediction is correct. Both outcomes are informative. Neither can be adjusted.

github.com/Eric-Robert-Lawson/attractor-oncology · doi.org/10.5281/zenodo.18986790 · ORCID: 0009-0002-0414-6544